Answer-Group Reranking for LLM Solution Selection

DASE7506 Mini-project 2 · Student 3036779750 · GitHub: byao128 · 26 October 2026

Abstract

This project studies whether comparing equivalent final answers helps select a correct solution from four supplied LLM candidates. A linear listwise scorer (B0) supplies candidate scores. The frozen G3 checkpoint groups candidates with the same parsed numeric answer, ranks groups with a small CatBoost model, and uses a 0.5/0.5 blend of B0 and tree group probabilities. In the primary train-only nested out-of-fold evaluation of the per-fold configuration-selection procedure, G3 selected 1,716/2,114 hard questions correctly versus 1,702 for the answer-group-aggregated B0 baseline; a second split/seed schedule gave a smaller gain of three questions. The frozen checkpoint scores 745/798 on validation and **381/467 (81.5846%) on the official test bank**, reproduced from the submission package. Its six inference assets occupy 1.743 MiB. A three-tree bagging extension failed its development gate. These results support a modest, split-sensitive selection gain, rather than verified mathematical reasoning. Multiple test evaluations and the subsequent submission preference are disclosed as an evaluation limitation.

1. Task and research questions

For every question the method returns one candidate index in {0,1,2,3}. The official parser and scorer compare the candidate's final numeric answer with the reference. Several candidates may share the correct answer, and any correct selection earns one point. Correct final answers do not establish that every intermediate reasoning step is valid.

The supplied bank has 4,755 training, 798 validation and 467 test questions, each with four candidate texts and a fixed 1,024-dimensional representation per candidate. References supervise training and development; inference uses only the supplied inputs. The course provides an empty selection interface, not a trained baseline. B0 and G3 in this report are project-trained methods, not instructor checkpoints.

Research question	Comparison used
Does comparing answers help?	B0 candidate argmax vs B0 answer-group aggregation
Does a learned group model add value?	B0 group vs tree branch vs G3 fusion
Does extra compute give stable gains?	Fixed single tree vs three-tree BAG3, same OOF folds

The central hypothesis is that group-level evidence can improve decisions when candidate answers disagree. The analysis therefore distinguishes the full bank from its **hard slice**: questions with at least two distinct valid parsed answers. Train has 2,114 hard questions and validation has 355. Every test question is hard by the course construction. This distinction matters when interpreting the validation-to-test gap.

2. Method: score candidates, compare answers

2.1 Candidate scorer and answer aggregation

B0 is a linear candidate scorer trained with a listwise objective: scores compete within each question, with probability mass assigned to correct candidates. Its fixed final recipe is 12 epochs, learning rate 0.003, L2 penalty 0.0001, batch size 128 and seed 2028. The implementation uses the supplied representations and text-derived features; the explicit candidate columns named valid and votes are excluded from B0 training. Parsing validity and answer support are still used by the group pipeline.

Let s(q,j) be the B0 score and G(q,g) the valid candidates in answer group g. The group logit is log(sum over j in G(q,g) of exp(s(q,j))). A softmax across valid groups gives pB(q,g). This pooling combines support for equivalent answers rather than treating their candidates as competing outputs. Invalid parses are excluded when valid candidates exist. The all-invalid fallback is candidate 0. Ties use the earliest candidate representative.

Inference stage	Information used / output
Four candidates to B0 scores	Candidate text features and fixed 1,024-D representations
Parser to answer groups	Numeric equivalence; no modified parser or reference access
Group features to tree scores	1,438 features; trained QuerySoftMax group ranker
Group probability fusion	p(q,g) = 0.5 pB(q,g) + 0.5 pT(q,g)
Winning group to candidate	Earliest representative of the highest-scoring group

2.2 Group representation and ranking

The 1,438 group features contain: mean raw embedding (1,024); mean, standard deviation, minimum and maximum of 64 PCA coordinates (256); group-minus-question and question-mean coordinates (128); mean/max of nine text features (18); and 12 support, cosine-similarity and B0-derived score/probability/margin features. PCA standardization and its 64 components are fit on training inputs only. No extra pretrained representation is downloaded.

The CatBoostRanker uses QuerySoftMax, so groups for one question compete within a query. Equivalent candidates have one group label, avoiding duplicate positive outputs. Trees are fit on hard training questions. The final tree has depth 4, L2 leaf regularization 10, 600 iterations, learning rate 0.03 and seed 2028. CPU threads are fixed to four. The saved configuration uses Plain boosting.

The method trains small selection models. It does not fine-tune the original LLM, generate new reasoning, call an API, or prove the arithmetic in a candidate. Its expected benefit is better comparison of available answers within the resource limits.

3. Development protocol and original G3 evidence

The original G3 study uses five outer training folds stratified by answer-group structure. Each training row is held out once. Inside each outer-training partition, a stratified 80/20 fit/selection split chooses the tree and fusion settings. B0 scores used to train the tree are produced by three-fold cross-fitting, so a row's training label does not train its own B0 feature score. PCA, B0, tree fitting and configuration selection for each outer evaluation remain inside its outer-training partition.

Selection dimension	Inner choices
Tree depth / leaf L2	{4, 6} / {10, 30}
Evaluated tree prefixes	{100, 200, 400, 600}
Tree mixture alpha	{0, 0.25, 0.5, 0.75, 1}
Selection order	Hard correct count, then group negative log likelihood; deterministic ties

Primary outer folds use seed 2026. The second run uses fold seed 2027 and a +10,000 training-seed offset, so it checks stability of the complete split/seed schedule, not solely the partition. Each outer fold selects its own configuration. The final frozen configuration is the modal full tuple among primary fold selections, with deterministic tie-breaking: depth 4, L2 10, 600 trees, alpha 0.5. Thus nested OOF evaluates the selection procedure, not exactly one fixed full-data checkpoint.

Original nested OOF	B0 group	G3	Hard net	Positive folds
2026: all / hard	4343/4755 1702/2114	4357/4755 1716/2114	+14	3/5
2027: all / hard	4339/4755 1698/2114	4342/4755 1701/2114	+3	3/5

The primary hard gain is +0.662 percentage points: 40 corrections and 26 regressions. The second run gives +0.142 points: 38 corrections and 35 regressions. The direction is positive in both, but the much smaller second gain indicates split/seed sensitivity. These runs do not establish a large or statistically robust generalization improvement.

Final checkpoint and validation record correction

The final B0 and tree use seed 2028; B0 training scores for the tree use three-fold split seed 6100. B0 and PCA are fit using all 4,755 training questions; tree supervision uses the 2,114 hard questions. Validation is not added to fitting. An early validation record reported 746/798 before CatBoost prediction explicitly passed thread_count=4. The four-thread repeat and serialized frozen bundle give **745/798**; that is the final result used throughout this report. The earlier number is retained only as diagnostic history.

4. Component analysis on the frozen validation bank

To explain the frozen model, five fixed methods were specified before this report analysis ran. The analysis is **retrospective and post-hoc**, not a new preregistered experiment. It fits no model or calibrator and changes neither the checkpoint nor its alpha. All methods use the same 798 validation inputs and original numeric parser. The hard slice has 355 questions.

Fixed method	All validation	Hard validation
Input majority	688/798 (86.216%)	245/355 (69.014%)
B0 candidate argmax	739/798 (92.607%)	296/355 (83.380%)
B0 answer-group (alpha=0)	742/798 (92.982%)	299/355 (84.225%)
Tree branch (alpha=1)*	743/798 (93.108%)	300/355 (84.507%)
G3 blend (alpha=0.5)	745/798 (93.358%)	302/355 (85.070%)

* Tree alpha=1 removes only the final B0 probability mixture. The tree still receives B0-derived features. It is not a model trained without B0.

Paired evidence: what changed on the same questions?

Improvement over comparator	Corrections	Regressions	Net
B0 candidate over input majority	75	24	+51
B0 group over B0 candidate	6	3	+3
G3 over B0 group	5	2	+3
G3 over tree alpha=1	4	2	+2

Learned candidate scores provide most of the improvement over majority on this bank. Pooling the same B0 scores by answer gives a further three correct questions. Adding the trained tree branch and equal-weight fusion gives another three. Fusion gains two questions over selecting the tree branch alone. These are small marginal gains; the 5 corrections and 2 regressions over B0 group deserve more attention than an accuracy rounded to a whole percentage.

The component controls isolate *output aggregation* and *final probability fusion*. They do not isolate the value of reasoning text versus embeddings, since both learned branches consume related input information and no new feature-removal model is fit. Consequently, the evidence supports answer-level comparison as a useful design choice here; it does not show that the model learned to verify every reasoning step.

Candidate-index disagreements can also exaggerate apparent behavioral differences. Two methods may return different candidates that have the same parsed final answer. The paired table therefore counts changes in correctness, not all changed indices. All four comparisons have identical net gains on the full and hard slices, because the other 443 validation questions are correctly selected by every method.

5. Confidence and the validation-to-test gap

For each method, confidence is the probability assigned to its selected candidate/group; majority uses the selected group's vote fraction. Calibration is assessed on the 355 hard validation questions using ten fixed equal-width bins. ECE is the bin-size-weighted absolute gap between mean confidence and accuracy. Binary Brier is mean squared error between selected-choice confidence and its correctness indicator; it is not a multiclass group Brier score.

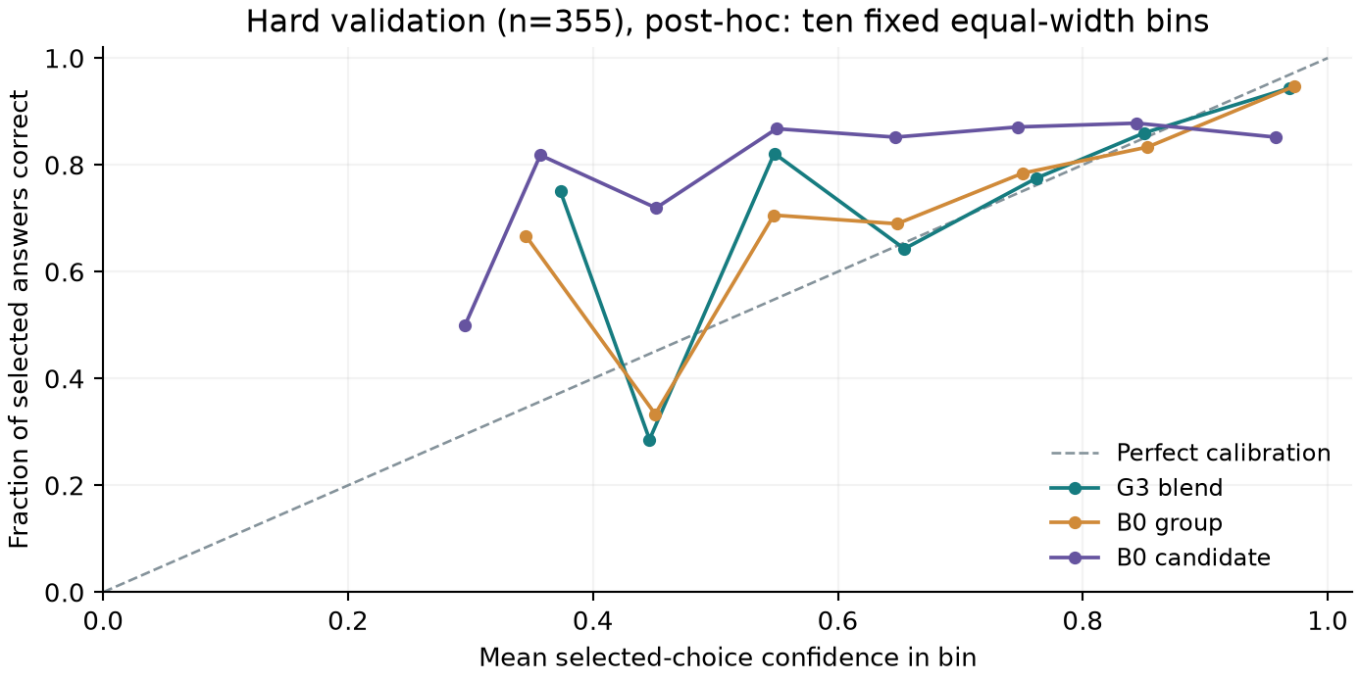


Figure 1. Nonempty bins only. Lines connect bin means and show no uncertainty intervals. Small bin counts can produce noisy movements. Candidate confidence is defined on a different output space from answer-group confidence.

Fixed method	ECE (lower better)	Binary Brier (lower better)
Input majority	0.0941	0.1930
B0 candidate argmax	0.2270	0.2027
B0 answer-group (alpha=0)	0.0445	0.1196
Tree branch (alpha=1)*	0.0442	0.1211
G3 blend (alpha=0.5)	0.0458	0.1182

G3 has slightly lower binary Brier than B0 group but slightly higher ECE. The tree branch has the lowest ECE here, despite lower accuracy than G3. Accuracy and calibration therefore do not rank the methods identically. B0 candidate scores can spread probability across several equivalent correct candidates; grouping changes the confidence event as well as the selected answer. These descriptive metrics do not establish calibrated posterior probabilities.

Full validation is 93.358%, but its 443 one-answer questions score 100%. Hard validation is **302/355 = 85.070%**, versus test **81.585%**. The apparent 11.774-point full-validation gap shrinks to 3.486 points using the more relevant hard slice. Different easy/hard composition explains the arithmetic of much of that gap. The remaining difference could reflect sample variation, further distribution differences or development selection bias. This evidence cannot distinguish their contributions.

6. Does more computation help? A negative result

A later experiment, BAG3, tested whether averaging three bootstrapped group-tree models could reduce instability. It retained the B0/PCA branch and the fixed G3 configuration, trained three hard-question bootstrap members, averaged their group probabilities, and mixed that average equally with B0 group probabilities. The comparison was made against a fixed single-tree G3 on the **same training OOF partition, seed 6061**. The declared improvement gate required at least ten additional hard correct answers and positive gains in at least three of five folds.

Separate BAG3 experiment (seed 6061)	All training OOF	Hard training OOF
B0 group	4337/4755	1696/2114
Fixed single G3	4345/4755	1704/2114
BAG3	4348/4755	1707/2114

BAG3 over fixed single G3	Observed
Corrections / regressions / net	24 / 21 / +3
Net changes across five folds	-1, +1, -1, -1, +5
Positive folds / gate result	2/5 / FAIL
Recorded external controller wall time	1,154.434 seconds

The pooled hard gain is only +0.142 percentage points. Three folds regress, and one fold accounts for most of the net gain. The extension fails both the net-improvement and positive-fold criteria. It therefore offers insufficient development evidence to replace the earlier method. More tree members can change errors without producing a reliable net benefit.

These numbers must not be combined with the original G3 nested OOF results. In particular, **1,707 vs 1,704** belongs to the fixed-configuration seed-6061 bagging comparison; **1,716 vs 1,702** belongs to the earlier nested selection procedure using seed 2026. Different fold/configuration protocols do not support a direct ranking of these counts.

Scope of the conclusion

The evidence rejects this particular three-member extension under the stated gate; it does not show that every ensemble method is ineffective. BAG3 increases the saved assets from 1,827,288 to 4,789,735 bytes (both below 16 MiB). A later full-bank BAG3 test was requested despite the failed development gate. Its result is recorded in the evaluation history on the next page, not used as evidence for a new method design. No BAG3 weights are submitted as the G3 checkpoint.

7. Official reproduction, resources and evaluation history

The submission package contains the original eight frozen inference sources and six checkpoint assets, with matching SHA-256 digests. On 26 October, a fixed-checkpoint reproduction launched the unchanged official prediction and scoring utilities as two fresh processes. It reproduced **381/467 = 81.5845824411135%**. No retraining or new model selection occurred in that reproduction. The scorer's raw JSON is retained unchanged, separately from resource metadata.

Resource / environment	Recorded value
Local CPU / OS	Intel Core i5-13500H / Windows 11
Python / NumPy / CatBoost	3.12.14 / 2.5.4 / 1.2.8
Fresh prediction / scoring process wall time	2.249177s / 0.465109s
External wrapper interval	2.748836s (includes both child processes)
Additional inference assets	1,827,288 bytes = 1.742638 MiB (six files)
Thread settings	OMP, OpenBLAS, MKL, NumExpr = 4; CatBoost = 4

Child processes run sequentially; their times include startup, imports, loading, preprocessing, inference and independent scoring. The wrapper interval also includes bookkeeping. The official scorer's internal scoring_seconds field is not the complete evaluation time. Local timing is evidence only: the course's 30-second eligibility is determined on its Xeon Platinum 8457C, which was not tested here.

Recorded wall costs include primary nested OOF 1,459.668s, stability OOF 2,009.366s and final bundle fitting 70.368s. They are individual logged runs, not a complete project cost. Total historical training/search cost is **unknown**. The old requirements file declared NumPy 2.5.3 and omitted CatBoost; the package instead pins the observed tested runtime, including NumPy 2.5.4 and CatBoost 1.2.8.

Test-use history and limitation

The retained records document original G3 scoring on 24 October (381/467), a later BAG3 evaluation on 26 October (378/467 = 80.9422%), and the fixed G3 reproduction above. Original G3 had already been retained on development evidence before the BAG3 test. However, the current preference to submit the older G3 followed comparison of aggregate test scores. This history **cannot be described as an untouched holdout used for a single final selection**. Test references are prohibited for selection by the guide. Instructor clarification is recommended before submission; no ruling is recorded. The technical reproduction checks do not establish policy compliance, and the disclosure is not a waiver of that rule.

8. Reproducibility, authorship and limitations

Reproduction and matching version

README.md gives exact installation, fixed train-only rebuilding, research experiment and fresh-process evaluation commands. The submitted checkpoint reproduces the score without training. records/reproduction/raw-score.json is the untouched scorer output; resources.json records the complete process timing. records/validation-analysis.json contains the fixed component/calibration aggregates, reproduced by code/validation_analysis.py. The original G3 OOF records and the separate BAG3 aggregate experiment record support the reported development comparisons.

The authoritative 14-file freeze is records/group-tree-v1/freeze.json. The whole package is covered by SHA256SUMS; VERSION.json identifies its local code-content hash. A public repository commit and matching checkpoint permalink must be recorded when the student publishes. No nonexistent Git commit or public submission URL is claimed here.

AI assistance and reused work

The student directed the project by setting the research goal and constraints, comparing proposed approaches, reviewing reported results, and deciding which experimental directions and model adjustments to pursue. OpenAI Codex assisted with discussing alternatives, implementing and revising code, debugging, automating experiments and integrity checks, and drafting the report and documentation. Later stages used GPT-5.6 Sol for complex implementation/review and GPT-6 Luna for routine execution; earlier stages also used other Codex models.

The parser, scorer, input format, data and fixed representations from the official course repository are reused unchanged. GSM8K questions/references retain their supplied MIT attribution. Candidate traces/representations originate from the recorded Qwen3-0.6B revision; no Qwen weights are included. NumPy and CatBoost provide numerical and ranking routines. No external training data or additional pretrained weights are introduced.

Conclusions and limits

Answer-group aggregation and a compact tree branch give small gains over the project's B0 on development data. The fixed checkpoint reproduces 81.5846% test accuracy within the locally measured size/time envelope. The bagging extension fails its development gate. The results remain limited by split/seed sensitivity, a small hard validation slice, reused validation, post-hoc analysis, unmeasured reference-machine runtime, incomplete total-cost records and the test-use history. They support candidate selection improvements on the supplied bank, not a general reasoning-verification capability.